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- Consider a symmetric Random Walk, that in each time unit is equally likely to take a unit step to the left or to the right.
- Now, speed up this process by taking smaller and smaller steps in smaller and smaller intervals.

What is Brownian Motion?

Formally, assume that every Δt time units we take a step of size Δx either to the left or to the right with equal probability.
Let $X(t)$ denote the position at time t. Then

$$X(t) = \Delta x \left(X_1 + X_2 + \cdots + X_{\lfloor t/\Delta t \rfloor} \right)$$

where

$$X_i = \begin{cases} +1 & \text{if the } i\text{th step of length } \Delta x \text{ is to the right} \\ -1 & \text{if it is to the left} \end{cases}$$

and the X_i are independent with

$$P(X_i = 1) = P(X_i = -1) = \frac{1}{2}.$$

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- $E[X(t)] = 0$
- $Var(X(t)) = (\Delta x)^2 \lfloor t/\Delta t \rfloor$
- Now we let $\Delta x = c\sqrt{\Delta t}$
- What we get is a normally distributed stochastic process $X(t)$ with mean 0 and variance $c^2 t$

Brownian Motion

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When $c = 1$, the process is called **Standard Brownian Motion**.

Sample Paths of Brownian Motion

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- But they are **nowhere differentiable** with probability 1

Thus Brownian motion paths are continuous but extremely rough - they are never smooth.

Brownian Motion is a Markov Process

Claim: Brownian motion $\{X(t), t \geq 0\}$ is a Markov process.

Proof:
The independent increments property implies that the increment

$$X(t+s) - X(s)$$

is independent of all process values before time s. Hence,

$$\begin{gathered} P(X(t+s) \leq a \mid X(s) = x, X(u), 0 \leq u < s) \\ = P(X(t+s) - X(s) \leq a - x \mid X(s) = x, X(u), 0 \leq u < s) \\ = P(X(t+s) - X(s) \leq a - x) \\ = P(X(t+s) \leq a \mid X(s) = x) \end{gathered}$$

Thus, the conditional distribution of the future state $X(t+s)$ given the present $X(s)$ and the past $\{X(u), 0 \leq u < s\}$ depends only on the present state.

Therefore, Brownian motion satisfies the **Markov property**.

Brownian Motion as a Gaussian Process

- A stochastic process $\{X(t), t \geq 0\}$ is a *Gaussian process* if

$$(X(t_1), X(t_2), \ldots, X(t_n))$$

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- Using independent increments,

$$f(x_1, \ldots, x_n) = f_{t_1}(x_1) f_{t_2 - t_1}(x_2 - x_1) \cdots f_{t_n - t_{n-1}}(x_n - x_{n-1})$$

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- Hence Brownian motion is a Gaussian process.

Covariance of Brownian Motion

Brownian motion can be characterized as a Gaussian process with

$$E[X(t)] = 0$$

and covariance

$$\mathrm{Cov}(X(s), X(t)) = s, \quad s \leq t$$

Idea:

$$X(t) = X(s) + (X(t) - X(s))$$

Since increments are independent,

$$\mathrm{Cov}(X(s), X(t) - X(s)) = 0$$

Thus

$$\mathrm{Cov}(X(s), X(t)) = \mathrm{Var}(X(s)) = s$$

Brownian Bridge

Suppose $\{X(t)\}$ is Brownian motion.
Define

$$Z(t) = X(t) - tX(1), \quad 0 \leq t \leq 1$$

Then

- $Z(0) = 0$
- $Z(1) = 0$

This process is called the **Brownian Bridge**.

Brownian Bridge Covariance

The Brownian Bridge is a Gaussian process with

$$E[Z(t)] = 0$$

and covariance

$$\mathrm{Cov}(Z(s), Z(t)) = s(1 - t), \quad s \leq t$$

Empirical Distribution Function

Let $X_1, X_2, \ldots, X_n$ be i.i.d. Uniform$(0,1)$.
Define

$$F_n(s) = \frac{N_n(s)}{n}$$

where

$$N_n(s) = \sum_{i=1}^{n} I(X_i \leq s)$$

Fluctuations of the Empirical Distribution

Define the scaled process

$$\alpha_n(s) = \sqrt{n}(F_n(s) - s)$$

Then as $n \to \infty$

$$\alpha_n(s) \Rightarrow Z(s)$$

where $Z(s)$ is a Brownian Bridge.

Key Takeaways

- Brownian motion is a Gaussian process with covariance

$$Cov(X(s), X(t)) = \min(s, t)$$

- Brownian Bridge = Brownian motion conditioned on $X(1) = 0$
- Covariance of Brownian bridge:

$$Cov(Z(s), Z(t)) = s(1 - t)$$

- Fluctuations of empirical distributions converge to a Brownian bridge

Hitting Time

- T_a is the first time a Brownian motion process reaches state a. That is, T_a is the minimum t such that, $X(t) = a$.
- We next attempt to compute $P(T_a \le t)$ by considering $P(X(t) \ge a)$.

$$\begin{aligned} P\{X(t) \ge a\} = {} & P\{X(t) \ge a \mid T_a \le t\} \, P\{T_a \le t\} \\ & + P\{X(t) \ge a \mid T_a > t\} \, P\{T_a > t\} \end{aligned}$$

Hitting Time

$$P\{X(t) \geq a \mid T_a \leq t\} \ = 1/2$$

- This is because, we reach state a at time T_a. $X(t) - X(T_a)$ is a gaussian distributed with mean 0.
- This implies that there is an equal probability of increase or decrease from a between times T_a and t.

$$P\{X(t) \geq a \mid T_a > t\} \ = 0$$

If $T_a > t$, that implies we have never reached a till time t. By continuity, this implies that $X(t) < a$.

Hitting Time

This implies,

$$P\{X(t) \geq a\} = \frac{1}{2} P\{T_a \leq t\}$$

$$\begin{aligned}
P\{T_a \leq t\} &= 2P\{X(t) \geq a\} \\
&= \frac{2}{\sqrt{2\pi t}} \int_a^\infty e^{-x^2/(2t)}\, dx \\
&= \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^\infty e^{-y^2/2}\, dy,
\end{aligned}$$

by setting $y = \dfrac{x}{\sqrt{t}}, \quad a > 0.$

Hitting Time for every state

$$P\{T_a < \infty\} = \lim_{t\to\infty} P\{T_a \leq t\} = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-y^2/2}\, dy = 1.$$

This implies that it is guaranteed to visit any state a in a Brownian Motion process.

Next, we'll try to compute $E[T_a]$.

Expected Value of Hitting Time

$$
\begin{aligned}
E[T_a] &= \int_0^\infty P\{T_a > t\}\, dt \qquad \text{[Tail Integral]} \\
&= \int_0^\infty \left(1 - \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^\infty e^{-y^2/2}\, dy\right) dt \qquad [1 - P\{T_a \leq t\}] \\
&= \frac{2}{\sqrt{2\pi}} \int_0^\infty \int_0^{a/\sqrt{t}} e^{-y^2/2}\, dy\, dt \\
&= \frac{2}{\sqrt{2\pi}} \int_0^\infty \int_0^{a^2/y^2} dt\, e^{-y^2/2}\, dy \\
&= \frac{2a^2}{\sqrt{2\pi}} \int_0^\infty \frac{1}{y^2} e^{-y^2/2}\, dy \\
&\geq \frac{2a^2 e^{-1/2}}{\sqrt{2\pi}} \int_0^1 \frac{1}{y^2}\, dy \\
&= \infty
\end{aligned}
$$

Hitting Time Final Remarks

- The expected value of hitting time being infinite with probability 1 of reaching a state implies that the walk is guaranteed to reach every state eventually but it might take infinite time to do so.
- Symmetrically, similar results hold when $a < 0$.

$$P\{T_a \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{|a|/\sqrt{t}}^{\infty} e^{-y^2/2}\, dy$$

Maximum Value

It is the maximum value a process takes upto time t.

$$
\begin{aligned}
P\left\{\max_{0\le s\le t} X(s) \ge a\right\} &= P\{T_a \le t\} \qquad \text{(by continuity)}\\
&= 2P\{X(t) \ge a\}\\
&= \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^{\infty} e^{-y^2/2}\, dy.
\end{aligned}
$$

Infinite Path Length

Given any time frame from t_1 to t_2, the length of the path traversed by the process is ∞!

$$S_n := \sum_{i=1}^{n} \left| B\left(\frac{iT}{n}\right) - B\left(\frac{(i-1)T}{n}\right) \right|.$$

For each i,

$$B\left(\frac{iT}{n}\right) - B\left(\frac{(i-1)T}{n}\right) \sim N\left(0, \frac{T}{n}\right),$$

and these increments are independent. Hence we may write

$$B\left(\frac{iT}{n}\right) - B\left(\frac{(i-1)T}{n}\right) \stackrel{d}{=} \sqrt{\frac{T}{n}}\, Z_i,$$

where $Z_1, \ldots, Z_n$ are i.i.d. $N(0,1)$. Therefore

$$S_n \stackrel{d}{=} \sqrt{\frac{T}{n}} \sum_{i=1}^{n} |Z_i|.$$

Now divide by $\sqrt{n}$:

$$\frac{S_n}{\sqrt{n}} \stackrel{d}{=} \sqrt{T}\,\frac{1}{n}\sum_{i=1}^{n}|Z_i|.$$

By the Strong Law of Large Numbers,

$$\frac{1}{n}\sum_{i=1}^{n}|Z_i| \to \mathbb{E}|Z_1| = \sqrt{\frac{2}{\pi}} \qquad \text{a.s.}$$

Hence

$$\frac{S_n}{\sqrt{n}} \to \sqrt{T}\,\sqrt{\frac{2}{\pi}} \qquad \text{a.s.}$$

Therefore,

$$S_n \sim \sqrt{\frac{2T}{\pi}}\,\sqrt{n} \qquad \text{a.s.}$$

and so

$$S_n \to \infty \qquad \text{a.s.}$$

Thus the sums

$$\sum_{i=1}^{n} \left| B\left(\frac{iT}{n}\right) - B\left(\frac{(i-1)T}{n}\right) \right|$$

grow without bound as $n \to \infty$.
This implies that the total path length is infinite.

As the granularity of the steps increases, the total path length increases. This goes towards infinity as the number of steps increases. The time frame length for all of these paths is equal.